Lecture 16: BLUE Estimator, and R^2 Coefficient, and Model Building

1 Best Linear Unbiased Estimator (BLUE)

The OLS $\hat{\boldsymbol{\beta}}$ is the best linear unbiased estimator (BLUE) of $\boldsymbol{\beta}$. The Gauss-Markov theorem states that, among all linear unbiased estimators of $\boldsymbol{\beta}$, the estimator $\hat{\boldsymbol{\beta}}$ is the one with the smallest variance. More formally, consider any other linear unbiased estimator $\tilde{\boldsymbol{\beta}}$ of $\boldsymbol{\beta}$, i.e. of the form $\tilde{\boldsymbol{\beta}} = \boldsymbol{A}y$ where $\boldsymbol{A}$ is a $p \times n$ non-stochastic matrix such that $\mathbb{E}\left[\tilde{\boldsymbol{\beta}}\right] = \boldsymbol{\beta}$. Then $Var(\tilde{\boldsymbol{\beta}}) \geq Var(\hat{\boldsymbol{\beta}})$, i.e., the matrix $Var(\tilde{\boldsymbol{\beta}}) - Var(\hat{\boldsymbol{\beta}})$ is positive semidefinite. Notice that the theorem does not require the assumptions that the vector $\boldsymbol{\epsilon}$ of error terms is Gaussian, but just that ϵ_i's are i.i.d. with zero mean and finite variance σ^2.

We do not have time to see the proof of this result. But, the interested student is referred to any standard statistics textbook (e.g. Mathematical Statistics and Data Analysis by John Rice).

2 The R^2 Coefficient

Suppose we have p explanatory variables $\boldsymbol{x}$. Suppose we observe $(\boldsymbol{x}_1^\top, y_1), \ldots, (\boldsymbol{x}_n^\top, y_n)$. Let the linear regression model $y = \boldsymbol{x}^\top \boldsymbol{\beta} + \epsilon$, and refer to it as the *full* model. Consider a reduced model $y = \boldsymbol{x}_a^T \boldsymbol{\beta}'$, where $\boldsymbol{x}_a$ stands for the first q explanatory variables in $\boldsymbol{x}$, i.e. $\boldsymbol{x}^\top = (\boldsymbol{x}_a^\top, \boldsymbol{x}_b^\top)$. Denote $\hat{\boldsymbol{y}}_{full}$ and $\hat{\boldsymbol{y}}_{rdcd}$ as the in-sample prediction under the full and reduced models, respectively. Moreover, denote by $\hat{\boldsymbol{\epsilon}}_{full}$ and $\hat{\boldsymbol{\epsilon}}_{rdcd}$ the residuals under the full and reduced models, respectively, i.e.,

$$\hat{\boldsymbol{\epsilon}}_{full} = \boldsymbol{y} - \hat{\boldsymbol{y}}_{full}, \qquad \hat{\boldsymbol{\epsilon}}_{rdcd} = \boldsymbol{y} - \hat{\boldsymbol{y}}_{rdcd}.$$

Then

$$\sum_{i=1}^{n} \hat{\epsilon}_{rdcd,i}^2 = \sum_{i=1}^{n} \left(\hat{y}_{rdcd,i} - \hat{y}_{full,i} + \hat{y}_{full,i} - y_i\right)^2 = \sum_{i=1}^{n} \left(\hat{y}_{rdcd,i} - \hat{y}_{full,i}\right)^2 + \sum_{i=1}^{n} \hat{\epsilon}_{full,i}^2. \tag{1}$$

Above, we have used the fact that the residual $\hat{\boldsymbol{\epsilon}}_{full}$ is orthogonal to the column space of X, and $\hat{y}_{full} - \hat{y}_{rdcd}$ is in the column space of X. Adding more explanatory variables always reduces the residual sum of squares. When $\boldsymbol{x}_a^\top = 1$, i.e., the constant is the only explanatory variable in the reduced model, then we have $\hat{y}_{rdcd,i} = \bar{y}$. In this case, $\sum_{i=1}^n \hat{\epsilon}_{rdcd,i}^2 = \sum_{i=1}^n (y_i - \bar{y})^2 =: S_{ttl}$ becomes the *total sum of squares*, and $\sum_{i=1}^n \left(\hat{y}_{rdcd,i} - \hat{y}_{full,i}\right)^2 = \sum_{i=1}^n \left(\hat{y}_i - \bar{y}\right)^2 =: S_{rgsn}$. This is referred to as the *regression sum of squares* of the full model: it is the sum of squares of the differences

between the fitted values and the average response variable. In other words, it measures how far the regression line is from $\bar{y}$. Recall that $S_{rsdl} := \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$. Hence, we can rewrite Eq. (1) as

$$S_{ttl} = S_{rgsn} + S_{rsdl}.$$

The R **squared** of a linear regression model is defined by

$$R^2 := \frac{S_{rgsn}}{S_{ttl}},$$

or equivalently

$$R^2 = 1 - \frac{S_{rsdl}}{S_{ttl}}.$$

Hence, $R^2 \leq 1$.
Question: What does it mean a high R^2?

Answer: It means that the model fits the data well. Indeed, R^2 gives the proportion of variability in $\boldsymbol{y}$ explained by the predictors. However, a high R^2 does not necessarily mean that the model is correct. Indeed, some explanatory variables can still be missing, and the prediction can thus be very poor. One can always increase R^2 by adding more explanatory variables, but this may lead to over-fitting, that is, basically, you would be fitting the noise as opposed to fitting the data. A small R^2 does not necessarily mean that the model is wrong: it may still be the case that the response variable is purely random. Because financial data are usually very noisy, we do not expect a high R^2. It is more important that the model has a high prediction power rather than a perfect in-sample fitting: this means that out-of-sample testing is important.

3 Model Building

Example: Suppose we want to test if (H_0) the reduced model $y = \beta_1 + \beta_2 x_2 + \epsilon$ is true, versus the hypothesis (H_1) the full model $y = \beta_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \epsilon$ is true. If either β_3 or β_4 are statistically different from zero, it means that the full model accounts better for variation in the data. Notice that, even if the reduced model is correct, we still have that the sum of square residuals of the full model is lower or equal than the corresponding sum in the reduced model.

3.1 Comparison of Full and Reduced Models

A random variable $X = \frac{U_1/d_1}{U_2/d_2}$ follows a *F-distribution* with degrees of freedom d_1 and d_2 if U_1 and U_2 have chi-squared distributions with d_1 and d_2 degrees of freedom, respectively, and U_1 and U_2 are independent. Suppose it is desired to test the null hypothesis that $\beta_{q+1} = \beta_{q+2} = \ldots = \beta_p = 0$ versus the alternative hypothesis that at least one of the predictors is not zero. Consider the statistic

$$F := \frac{\frac{S_{rdcd,rsdl} - S_{full,rsdl}}{p-q}}{\frac{S_{full,rsdl}}{n-p}},$$

where $S_{rdcd,rsdl}$ and $S_{full,rsdl}$ are the residual sums of squares of the reduced and full models, respectively. Under the null hypothesis, and assuming that the error terms in the linear regression

model are Gaussian i.i.d. random variables, this statistics follows a F distribution with degrees of freedom $p - q$ and $n - p$.

The test rejects the null hypothesis when F is sufficiently large. The F statistics is also at the heart of sequential techniques to decide on the most suitable linear model. You first collect variables that have some plausible relationship with the response variable. After the data collection, you decide on a final model using one of these two popular procedures.

- **Forward Stepwise Selection:** start from a small set of explanatory variables. Consider all models obtained by adding one more predictor.

  1. For each variable not already included, calculate the *F-to-enter*, that is the value of the F statistic when this additional variable is included in the model. You then select the variable with the largest *F-to-enter*.
  2. If the largest F-to-enter is greater than some specified critical value F_{in}, then add the predictor to the model (typical choices are $2 \leq F_{in} \leq 4$).
  3. Repeat the two steps above until no predictors can be added.

An alternative procedure is

- **Backward Stepwise Elimination:** Start with the full model, that is inclusive of all collected predictor variables. Consider all models obtained by deleting one predictor.

  1. For each variable, calculate the *F-to-remove*, that is the value of the F statistics when this variable is removed. You then select the variable with the smallest *F-to-remove*.
  2. If the smallest F-to-remove is smaller than some specified critical value $2 \leq F_{out} \leq 4$, then remove the predictor variable from the model.
  3. Repeat the two steps above until no predictors can be removed.

A $F = t^2$

We show that, in a simple linear regression, we have that F statistics is equal to the square of the t statistics. To show this, we first prove that $R^2 = \rho^2$ in a simple linear regression.

We begin by defining some key quantities in simple linear regression. Let the observed values be (x_i, y_i) for $i = 1, \ldots, n$. Then:

- Total Sum of Squares (TSS):

$$\text{TSS} = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \text{SS}_{yy}$$

- Residual Sum of Squares (RSS or SSR):

$$\text{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

We know that in simple linear regression:

$$\hat{y}_i = \hat{\alpha} + \hat{\beta} x_i$$

Using the least squares estimators, we have

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}, \quad \hat{\beta} = \frac{\mathrm{SS}_{xy}}{\mathrm{SS}_{xx}}$$

So we can write:

$$\hat{y}_i = \bar{y} + \hat{\beta}(x_i - \bar{x})$$

Then the residual sum of squares can be expressed as:

$$\mathrm{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} \Big[(y_i - \bar{y}) - \hat{\beta}(x_i - \bar{x}) \Big]^2$$

Expanding the square:

$$\mathrm{RSS} = \sum_{i=1}^{n} (y_i - \bar{y})^2 + \hat{\beta}^2 \sum_{i=1}^{n} (x_i - \bar{x})^2 - 2\hat{\beta} \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})$$

Using notation:

$$\mathrm{SS}_{yy} = \sum (y_i - \bar{y})^2, \quad \mathrm{SS}_{xx} = \sum (x_i - \bar{x})^2, \quad \mathrm{SS}_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})$$

We get:

$$\mathrm{RSS} = \mathrm{SS}_{yy} + \hat{\beta}^2 \mathrm{SS}_{xx} - 2\hat{\beta}\mathrm{SS}_{xy}$$

Substitute $\hat{\beta} = \frac{\mathrm{SS}_{xy}}{\mathrm{SS}_{xx}}$:

$$\mathrm{RSS} = \mathrm{SS}_{yy} + \left(\frac{\mathrm{SS}_{xy}^2}{\mathrm{SS}_{xx}} \right) - 2 \left(\frac{\mathrm{SS}_{xy}^2}{\mathrm{SS}_{xx}} \right) = \mathrm{SS}_{yy} - \frac{\mathrm{SS}_{xy}^2}{\mathrm{SS}_{xx}}$$

Then we have:

$$R^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}} = 1 - \frac{\mathrm{SS}_{yy} - \frac{\mathrm{SS}_{xy}^2}{\mathrm{SS}_{xx}}}{\mathrm{SS}_{yy}} = \frac{\mathrm{SS}_{xy}^2}{\mathrm{SS}_{xx} \cdot \mathrm{SS}_{yy}}$$

Now recall the definition of Pearson correlation coefficient:

$$\rho = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X) \cdot \mathrm{Var}(Y)}} = \frac{\mathrm{SS}_{xy}}{\sqrt{\mathrm{SS}_{xx} \cdot \mathrm{SS}_{yy}}} \Rightarrow \rho^2 = \frac{\mathrm{SS}_{xy}^2}{\mathrm{SS}_{xx} \cdot \mathrm{SS}_{yy}}$$

Hence, we conclude:

$$R^2 = \rho^2$$

Proof that $F = t^2$ in Simple Linear Regression

From the previous derivation, we know that:

$$\text{RSS} = \text{SS}_{yy} - \frac{\text{SS}_{xy}^2}{\text{SS}_{xx}} \qquad (\star)$$

We now prove that in simple linear regression, the F-statistic is equal to the square of the t-statistic when testing the null hypothesis $H_0 : \beta = 0$.

In the general linear model, the F-statistic is given by:

$$F = \frac{(\text{TSS} - \text{RSS})/(p - q)}{\text{RSS}/(n - p)}$$

where:

- p is the number of parameters in the full model ($p = 2$ in SLR),
- q is the number of parameters in the restricted model ($q = 1$ in SLR),
- n is the number of observations.

So in simple linear regression:

$$F = \frac{(\text{SS}_{yy} - \text{RSS})/1}{\text{RSS}/(n - 2)} = \frac{(n - 2)(\text{SS}_{yy} - \text{RSS})}{\text{RSS}}$$

The t-statistic for testing $H_0 : \beta = 0$ is given by:

$$t = \frac{\hat{\beta} - 0}{\text{SE}(\hat{\beta})}$$

$$t^2 = \frac{\hat{\beta}^2}{\text{Var}(\hat{\beta})}$$

We know:

$$\hat{\beta} = \frac{\text{SS}_{xy}}{\text{SS}_{xx}}, \quad \text{Var}(\hat{\beta}) = \frac{\hat{\sigma}^2}{\text{SS}_{xx}}, \quad \hat{\sigma}^2 = \frac{\text{RSS}}{n - 2}$$

So:

$$t^2 = \frac{\left(\frac{\text{SS}_{xy}}{\text{SS}_{xx}}\right)^2}{\frac{\text{RSS}}{(n-2)\text{SS}_{xx}}} = \frac{\text{SS}_{xy}^2}{\text{SS}_{xx}} \cdot \frac{(n - 2)}{\text{RSS}}$$

Now use Equation $\star$:

$$\frac{\mathrm{SS}_{xy}^2}{\mathrm{SS}_{xx}} = \mathrm{SS}_{yy} - \mathrm{RSS} \Rightarrow t^2 = \frac{(n-2)(\mathrm{SS}_{yy} - \mathrm{RSS})}{\mathrm{RSS}} = F$$

$$F = t^2$$

A.1 Relation between F-statistic and t-statistics in MLR:

The F-statistic tests the joint hypothesis

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0.$$

It is computed as:

$$F = \frac{(\mathrm{RSS}_{\mathrm{reduced}} - \mathrm{RSS}_{\mathrm{full}})/k}{\mathrm{RSS}_{\mathrm{full}}/(n-k-1)}.$$

For testing whether a single predictor $H_0 : \beta_j = 0$, the corresponding t-statistic is:

$$t_j = \frac{\hat{\beta}_j}{\mathrm{SE}(\hat{\beta}_j)}.$$

The F-statistic involves **all** k predictors simultaneously. The t-statistic tests **only one** predictor at a time, conditioning on all others being in the model. Thus, $F = t^2$ holds in MLR **only** if the F-test tests a single restriction (i.e., $H_0 : \beta_j = 0$ for some fixed j), in which case the test is equivalent to the corresponding two-sided t-test.

If $k > 1$ and the F-test involves multiple restrictions, the test evaluates a **joint null hypothesis** and aggregates evidence across predictors — hence, $F \neq t^2$ in general.